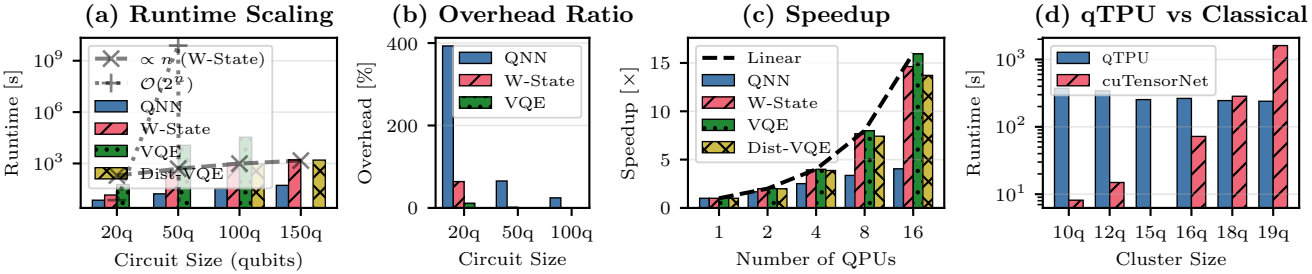
(a) Runtime Scaling
Runtime [s]
10^9
10^6
10^3
$\propto n$ (W-State)
$\mathcal{O}(2^n)$
QNN
W-State
VQE
Dist-VQE
20q
50q
100q
150q
Circuit Size (qubits)
(b) Overhead Ratio
Overhead [%]
400
200
0
QNN
W-State
VQE
20q
50q
100q
Circuit Size
(c) Speedup
Speedup [×]
15
10
5
0
Linear
QNN
W-State
VQE
Dist-VQE
1
2
4
8
16
Number of QPUs
(d) qTPU vs Classical
Runtime [s]
10^3
10^2
10^1
qTPU
cuTensorNet
10q
12q
15q
16q
18q
19q
Cluster Size